



Savitribai Phule Pune University Fourth Year of Computer Engineering (2019 Course) Elective V 410252(A): Natural Language Processing		
Teaching Scheme:	Credit	Examination Scheme:
TH: 03 Hours/Week	03	In-Sem (Paper): 30 Marks End-Sem (Paper): 70 Marks
Prerequisite Courses: Discrete Mathematics (210241), Theory of Computation (310242), Data Science and Big Data Analytics (310251)		
Companion Course: Laboratory Practice VI(410255)		
Course Objectives: - To be familiar with fundamental concepts and techniques of natural language processing (NLP) - To acquire the knowledge of various morphological, syntactic, and semantic NLP tasks - To develop the various language modeling techniques for NLP - To use appropriate tools and techniques for processing natural languages - To comprehend the advance real world applications in NLP domain. - To Describe Applications of NLP and Machine Translations.		
Course Outcomes: On completion of the course, student will be able to– CO1: Describe the fundamental concepts of NLP, challenges and issues in NLP CO2: Analyze Natural languages morphologically, syntactical and semantically OR Describe the concepts of morphology, syntax, semantics of natural language CO3: Illustrate various language modelling techniques CO4: Integrate the NLP techniques for the information retrieval task CO5: Demonstrate the use of NLP tools and techniques for text-based processing of natural languages CO6: Develop real world NLP applications		
Course Contents		
Unit I	Introduction to Natural Language Processing	07 Hours
Introduction: Natural Language Processing, Why NLP is hard? Programming languages Vs Natural Languages, Are natural languages regular? Finite automata for NLP, Stages of NLP, Challenges and Issues(Open Problems) in NLP Basics of text processing: Tokenization, Stemming, Lemmatization, Part of Speech Tagging		
#Exemplar/Case Studies	Why English is not a regular language: http://cs.haifa.ac.il/~shuly/teaching/08/nlp/complexity.pdf#page=20	
*Mapping of Course Outcomes for Unit I	CO1	
Unit II	Language Syntax and Semantics	07 Hours

Morphological Analysis: What is Morphology? Types of Morphemes, Inflectional morphology & Derivational morphology, Morphological parsing with Finite State Transducers (FST) Syntactic Analysis: Syntactic Representations of Natural Language, Parsing Algorithms, Probabilistic context-free grammars, and Statistical parsing Semantic Analysis: Lexical Semantic, Relations among lexemes & their senses – Homonymy, Polysemy, Synonymy, Hyponymy, WordNet, Word Sense Disambiguation (WSD), Dictionary based approach, Latent Semantic Analysis		
<u>#Exemplar/Case Studies</u>	Study of Stanford Parser and POS Tagger https://nlp.stanford.edu/software/lex-parser.html https://nlp.stanford.edu/software/tagger.html	
<u>*Mapping of Course Outcomes for Unit II</u>	CO2	
Unit III	Language Modelling	07 Hours
Probabilistic language modeling, Markov models, Generative models of language, Log-Linear Models, Graph-based Models N-gram models: Simple n-gram models, Estimation parameters and smoothing, Evaluating language models, Word Embeddings/ Vector Semantics: Bag-of-words, TFIDF, word2vec, doc2vec, Contextualized representations (BERT) Topic Modelling: Latent Dirichlet Allocation (LDA), Latent Semantic Analysis, Non Negative Matrix Factorization		
<u>#Exemplar/Case Studies</u>	Study of language modelling for Indian languages.	
<u>*Mapping of Course Outcomes for Unit III</u>	CO3	
Unit IV	Information Retrieval using NLP	07 Hours
Information Retrieval: Introduction, Vector Space Model Named Entity Recognition: NER System Building Process, Evaluating NER System Entity Extraction, Relation Extraction, Reference Resolution, Coreference resolution, Cross Lingual Information Retrieval		
<u>#Exemplar/Case Studies</u>	Natural Language Processing based Information Extraction & Retrieval: https://www.cdac.in/index.aspx?id=mc_cli_cross_lingual_info	
<u>*Mapping of Course Outcomes for Unit IV</u>	CO4	
Unit V	NLP Tools and Techniques	08 Hours
Prominent NLP Libraries: Natural Language Tool Kit (NLTK), spaCy, TextBlob, Gensim etc. Linguistic Resources: Lexical Knowledge Networks, WordNets, Indian Language WordNet (IndoWordnet), VerbNets, PropBank, Treebanks, Universal Dependency Treebanks Word Sense Disambiguation: Lesk Algorithm Walker's algorithm, WordNets for Word Sense Disambiguation		
<u>#Exemplar/Case Studies</u>	Hindi Wordnet: https://www.cfilt.iitb.ac.in/wordnet/webhwn/ Sanskrit WordNet: https://www.cfilt.iitb.ac.in/wordnet/webswn/ Indic Library: http://anoopkunchukuttan.github.io/indic_nlp_library/	

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<u>*Mapping of Course Outcomes for Unit V</u>	CO5											
Unit VI	Applications of NLP	07 Hours										
Machine Translation: Rule based techniques, Statistical Machine Translation (SMT), Cross Lingual Translation												
Sentiment Analysis, Question Answering, Text Entailment, Discourse Processing, Dialog and Conversational Agents, Natural Language Generation												
<u>#Exemplar/Case Studies</u>	Study working of Google Translate Study working of IBM Watson Natural Language Processing											
<u>*Mapping of Course Outcomes for Unit VI</u>	CO6											
Learning Resources												
Text Books:												
1. Jurafsky, David, and James H. Martin, “Speech and Language Processing: An Introduction to Natural Language Processing”, Computational Linguistics and Speech Recognition, PEARSON Publication 2. Manning, Christopher D., and rich Schütze, “Foundations of Statistical Natural Language Processing”, Cambridge, MA: MIT Press												
Reference Books:												
1. Steven Bird, Ewan Klein, Edward Loper, “Natural Language Processing with Python – Analyzing Text with the Natural Language Toolkit”, O’Reilly Publication 2. Dipanjan Sarkar, “Text Analytics with Python: A Practical Real-World Approach to Gaining Actionable Insights from your Data”, Apress Publication ISBN: 9781484223871 3. Alexander Clark, Chris Fox, and Shalom Lappin, “The Handbook of Computational Linguistics and Natural Language Processing”, Wiley Blackwell Publications 4. Jacob Eisenstein, “Natural Language Processing”, MIT Press 5. Jacob Eisenstein, “An Introduction to Information Retrieval”, Cambridge University Press												
e-Books :												
1. https://web.stanford.edu/~jurafsky/slp3/ed3book.pdf 2. https://www3.cs.stonybrook.edu/~cse521/L16NLP.pdf												
NPTEL Courses links:												
• https://nptel.ac.in/courses/106101007 • https://nptel.ac.in/courses/106106211												
<u>@The CO-PO Mapping Matrix</u>												
CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	-	-	-	-	-	-	-	-	-
CO2	3	3	2	2	2	-	-	-	-	-	-	1
CO3	2	3	3	2	2	-	-	-	-	-	-	2
CO4	2	2	3	3	3	-	2	2	-	-	-	3
CO5	2	2	3	3	3	-	-	-	-	-	-	3
CO6	3	3	3	3	3	2	1	1	-	-	-	3